

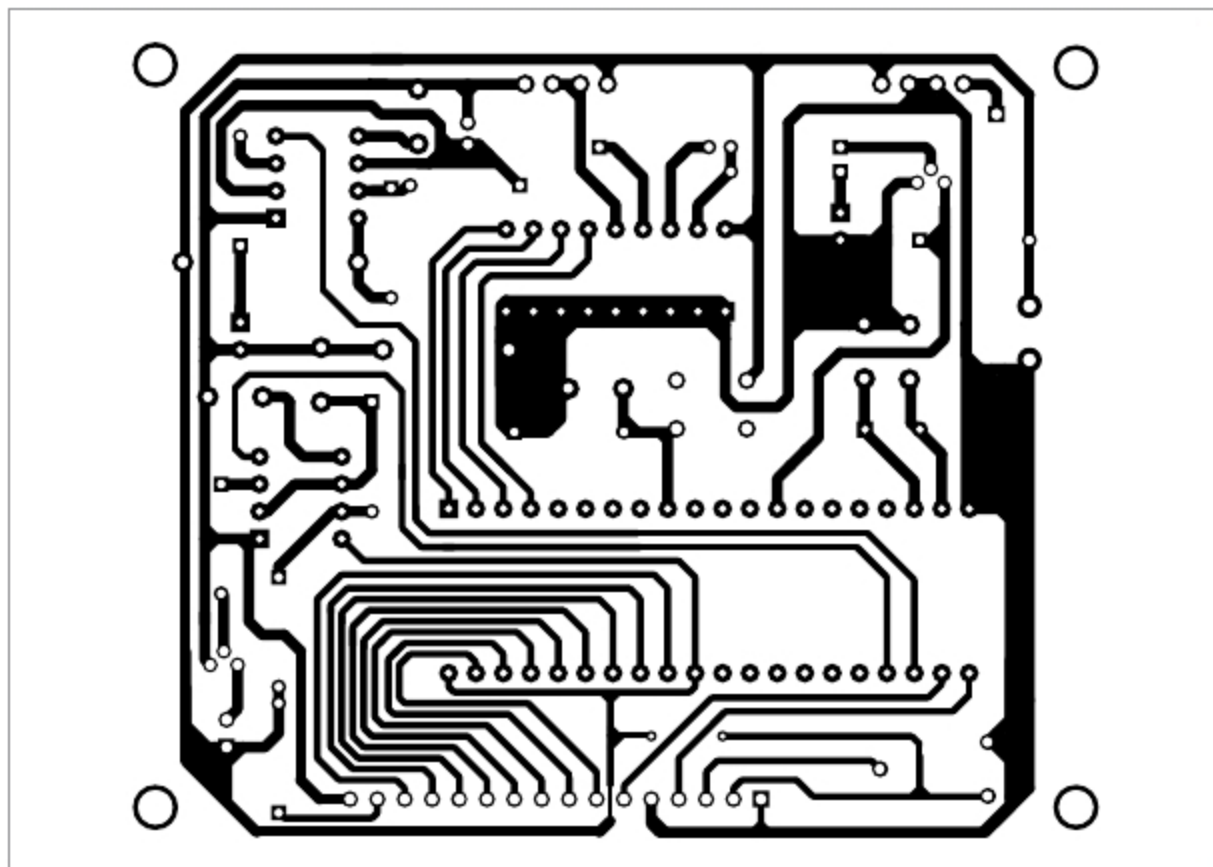


Fig. 7: PCB layout of receiver unit

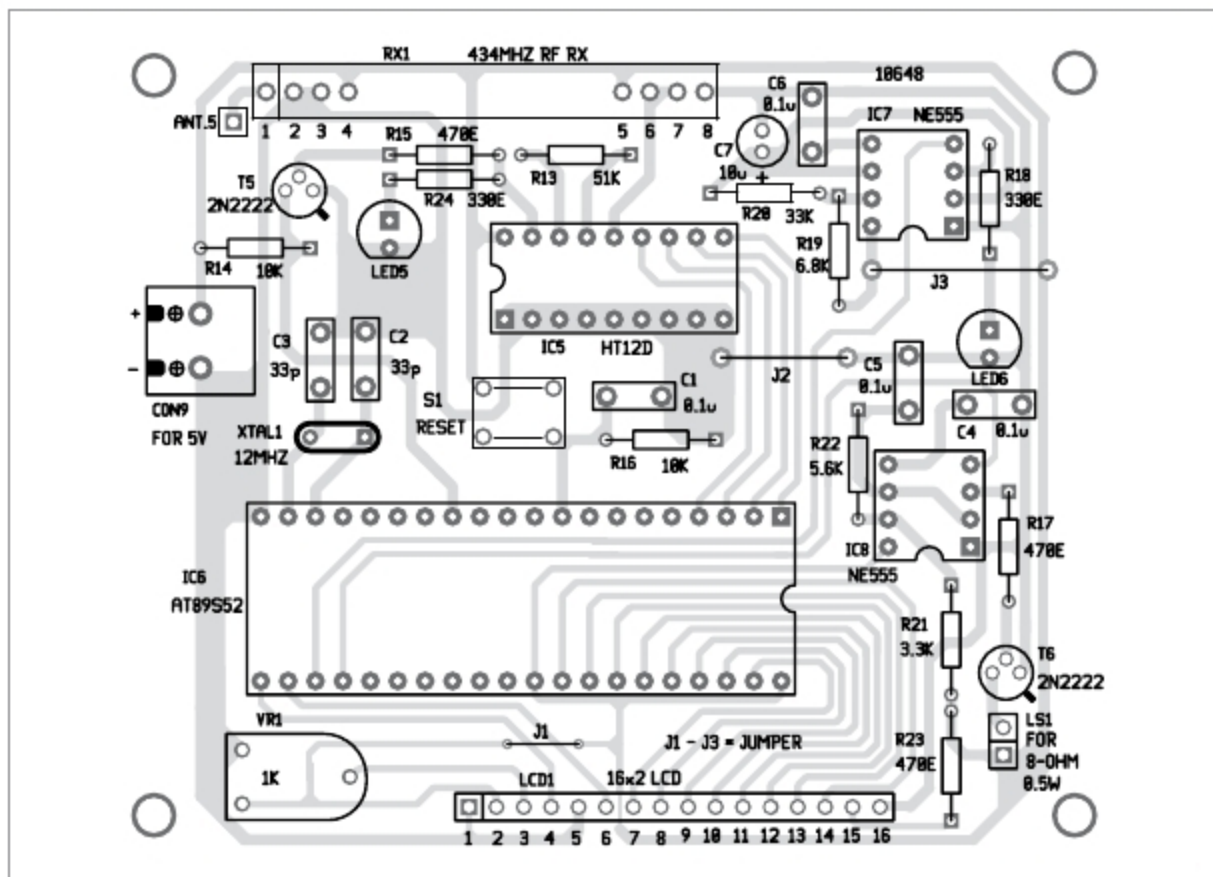


Fig. 8: Components layout for receiver PCB

ext_int() is an interrupt sub-routine function that gets input from port P1

delay_10s() generates approximately 10-sec delay

main() compares digital input of P1 and detects which port pin is low. It also displays on LCD1 the side of motion or intrusion detection

Construction and testing

An actual-size PCB layout for the transmitter section of the wireless

security system is shown in Fig. 5 and its components layout in Fig. 6.

Cut the PCB across the dotted lines for four transmitter sets. Enclose each transmitter section in separate boxes, including 6V battery. Connect the four

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

PARTS LIST

Semiconductors:

IC1-IC4	- HT12E encoder
IC5	- HT12D decoder
IC6	- AT89S52/AT89C52 MCU
IC7, IC8	- NE555 timer
T1-T6	- 2N2222 npn transistor
LED1-LED6	- 5mm LED
D1-D8	- 1N4007 rectifier diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1, R4, R7, R10	- 1-mega-ohm
R2, R5, R8, R11	- 680-ohm
R3, R6, R9, R12,	
R15, R17, R23	- 470-ohm
R13	- 51-kilo-ohm
R14, R16	- 10-kilo-ohm
R18	- 330-ohm
R19	- 6.8-kilo-ohm
R20	- 33-kilo-ohm
R21	- 3.3-kilo-ohm
R22	- 5.6-kilo-ohm
R24	- 330-ohm
VR1	- 1-kilo-ohm preset

Capacitors:

C1, C4-C6 - 0.1 μ F ceramic disk
C2, C3 - 33pF ceramic disk
C7 - 10 μ F, 16V electrolytic

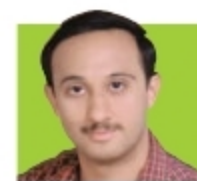
Miscellaneous:

- CON1, CON3, CON5, CON7, CON9 - 2-pin terminal connector
- CON2, CON4, CON6, CON8 - 3-pin connector
- LCD1 - 16x2 line LCD
- S1 - Tactile switch
- LS1 - 8-ohm, 0.5W speaker
- PS1-PS4 - PIR sensor (HC-SR501)
- XTAL1 - 12MHz crystal
- TX1-TX4 - 434MHz RF transmitter module
- RX1 - 434MHz RF receiver module
- ANT.1-ANT.5 - Antenna wire up to 17cm long
 - IC bases
 - 5V DC regulated power supply
- BATT.1-BATT.4 - 6V battery for each transmitter

PIR sensors in each transmitter PCB assembly. Install these at the four locations of your home or office.

An actual-size PCB layout of the receiver unit is shown in Fig. 7 and its components layout in Fig. 8.

After assembling the circuits on the PCBs, if everything is correct, burn the program (pir.hex) into the MCU using a suitable programmer. Insert AT89S52 on IC base soldered on the PCB. Enclose the PCB, switch and connectors in a suitable box. Connect 5V to operate the circuit. Your circuit is now ready to use. **EFY**



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